Volume of Rectangular Prisms

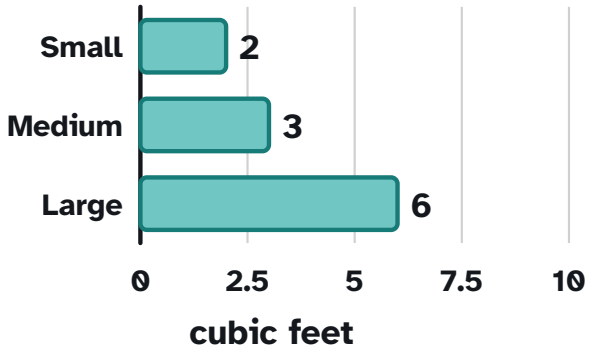
Find the volume of several boxes, with exact decimal edges, and use the volumes to choose.

Name _____ **Date** _____

Tonight's spark

A class needs a storage box that holds at least 5 cubic feet. Small is 2 × 1 × 1 ft. Medium is 2 × 1.5 × 1 ft. Large is 2 × 2 × 1.5 ft. Which box works?

Guess first: does the Medium box hold enough?



Our class model: exact edges, then compare

Try it

Work it out on paper. Write one answer on each line.

1 **What is the volume of a box that is 3 m by 2 m by 4 m?**

A. 9 m^3 **B.** 24 m^3 **C.** 14 m^3

My answer _____

2 **A box is 2 ft by 5 ft by 1.5 ft. What is its volume?**

A. 15 ft^3 **B.** 10 ft^3 **C.** 20 ft^3 **D.** 8.5 ft^3

My answer _____

3 **Box P is 3 by 2 by 1.5. Box Q is 2 by 2 by 2. Which holds more?**

A. Box Q, because 8 is greater than 6 **B.** They hold the same

C. Box P, 9 cubic units to 8 cubic units **D.** Box Q, because it has no fractions

My answer _____

4 **A shop needs a box that holds MORE than 20 cubic feet. Which box works?**

A. 4 × 2 × 2 **B.** 3 × 3 × 2 **C.** 2 × 2 × 5 **D.** 5 × 2 × 2.5

My answer _____

5 **Box M is 6 by 3 by 1.5. Box N is 5 by 4 by 1. Which holds more, and by how much?**

A. Box N, by 7 cubic units **B.** Box M, by 7 cubic units **C.** Box M, by 1 cubic unit

D. Box N, because its base is bigger

My answer _____

Talk about it

Which box holds more, and how much more does it hold?

Home mission

Pick two boxes, like a cereal box and a snack box. Measure each edge to the nearest half inch. Find each volume without rounding. Which holds more?

If we get stuck, ask

- Did you use the exact edge, or did you round it?
- How many edges go into a volume?
- What number does the problem say we have to reach?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/5-10-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it